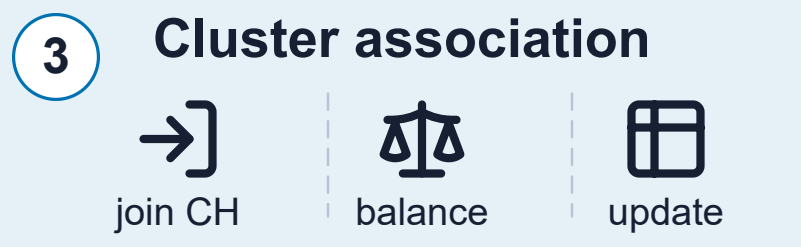
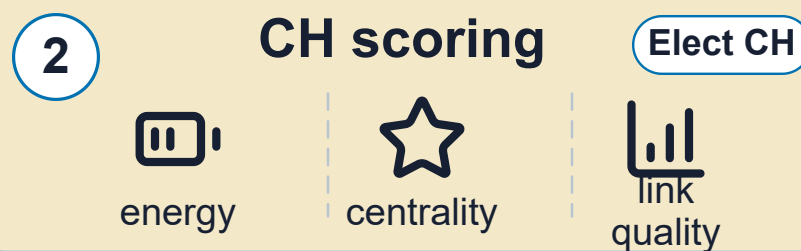
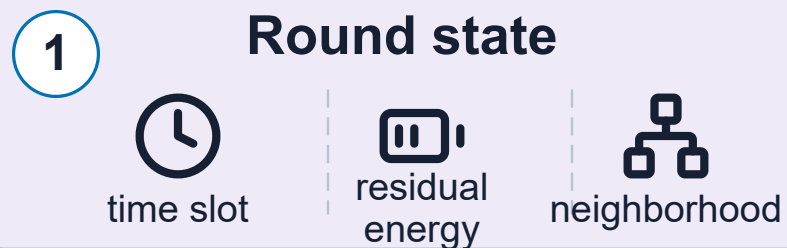
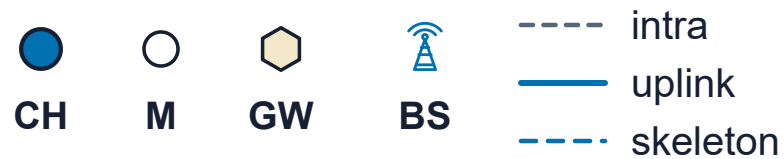
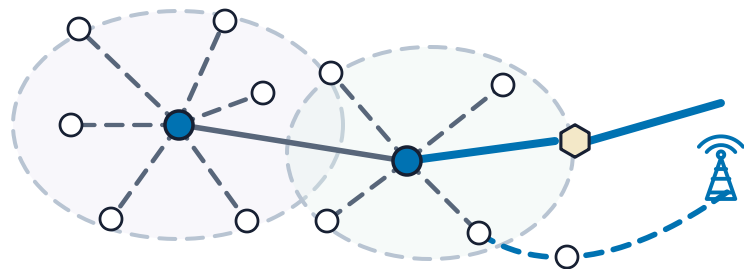




Phase 1 Cluster Formation



Phase 2 Intra-Cluster Communication

Inputs: energy · link quality · density · buffer

1 CAS mode selector

select one mode

Direct

$M \rightarrow CH$



Chain

$M \rightarrow M \rightarrow CH$

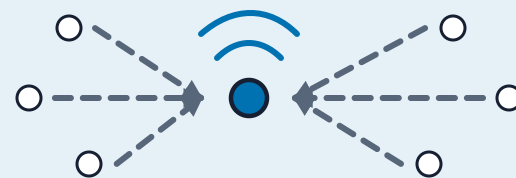


Two-hop

$M \rightarrow R \rightarrow CH$



2 Intra-cluster forwarding



3 CH aggregation



deduplicate



fuse



buffer



Phase 3 Uplink to BS

pre-transmission route selection

